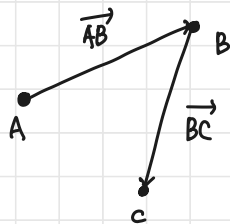


위치벡터

벡터 : 시점이 자유로움



$$\vec{AB} = \vec{OB} - \vec{OA}$$

위치벡터 : 시점이 원점 O으로 고정

$$\vec{a} = (2, 3) \quad \vec{b} = (4, 6)$$

☆ 내분점 / 외분점

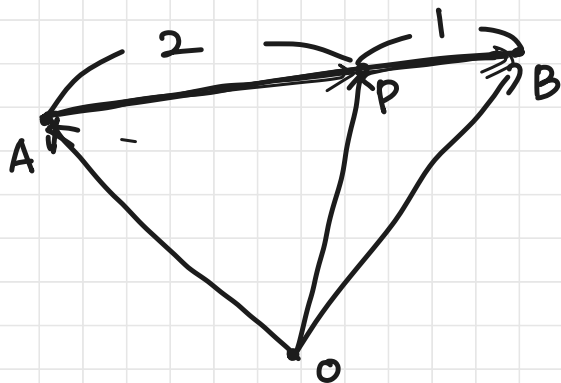


$$\overline{AP} : \overline{PB} = m : n$$



$$\overline{AQ} : \overline{QB} = m : n$$

#4/6-11.



$\vec{OP}$

$$\overline{AP} : \overline{PB} = 2 : 1$$

$$2\overline{PB} = \overline{AP}$$

$$\vec{AP} = \frac{1}{2}\vec{PB} = \left(\frac{2}{3}\vec{AB}\right)$$

$$\vec{AP} = \vec{OP} - \vec{OA}$$

$$\frac{2}{3}\vec{AB} = \frac{2}{3}[\vec{OB} - \vec{OA}]$$

$$\vec{OP} - \vec{OA} = \frac{2}{3}(\vec{OB} - \vec{OA})$$

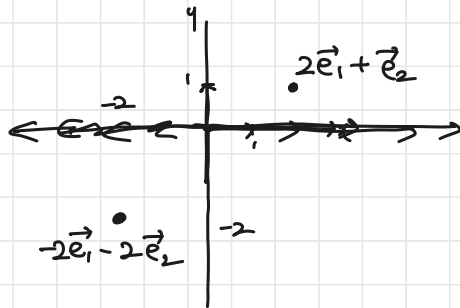
$$\vec{OP} = \frac{2}{3}\vec{OB} - \frac{2}{3}\vec{OA} + \vec{OA}$$

$$= \frac{2}{3}\vec{OB} + \frac{1}{3}\vec{OA}$$

표준 정규 기저 : 2차 평면 (2D)
(1) (basis)

$$\vec{e}_1 = (1, 0)$$

$$\vec{e}_2 = (0, 1)$$



• 3차 공간 (3D)

$$\vec{e}_1 = (1, 0, 0)$$

$$\vec{e}_2 = (0, 1, 0)$$

$$\vec{e}_3 = (0, 0, 1)$$

벡터의 성분

$$\vec{a} = (a_1, a_2) = a_1 \vec{e}_1 + a_2 \vec{e}_2$$

$$\vec{a} = (a_1, a_2, a_3) = a_1 \vec{e}_1 + a_2 \vec{e}_2 + a_3 \vec{e}_3$$

$\underbrace{\hspace{1cm}}_{\text{벡터 } \vec{a}(\vec{a}) \text{의 성분}} \quad \begin{matrix} \swarrow \text{1차 성분} & \searrow \text{2차 성분} \\ \end{matrix}$

 $\underbrace{\hspace{1cm}}_{\text{2차 성분}}$

벡터의 크기 & 연산

(1) 2차원 평면에서 $\vec{a} = (a_1, a_2)$, $\vec{b} = (b_1, b_2)$

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2} \quad \vec{a} \pm \vec{b} = (a_1 \pm b_1, a_2 \pm b_2) \quad k\vec{a} = k(a_1, a_2) = (ka_1, ka_2)$$

$$(kb_1, kb_2, kb_3)$$

(2) 3차원 공간에서 $\vec{a} = (a_1, a_2, a_3)$, $\vec{b} = (b_1, b_2, b_3)$

$$|\vec{b}| = \sqrt{b_1^2 + b_2^2 + b_3^2} \quad \vec{a} \pm \vec{b} = (a_1 \pm b_1, a_2 \pm b_2, a_3 \pm b_3) \quad k\vec{b} = k(b_1, b_2, b_3)$$

#예 6-12.

$$\vec{a} = (2p-3, 4) \quad \vec{b} = (1, -3q+1)$$

$$\begin{cases} 2p-3=1 \\ 4 = -3q+1 \end{cases} \Rightarrow \begin{cases} 2p=4 \Rightarrow p=2 \\ -3q=3 \Rightarrow q=-1 \end{cases}$$

#예 6-13.

$$\begin{aligned} (3, 16, -12) &= m(0, 2, -3) + n(1, 4, -2) \\ &= (0, \underline{2m}, -3m) + (n, \underline{4n}, -2n) \end{aligned}$$

$$n=3$$

$$2m + \textcircled{4n} = 16$$

12

$$m=2$$

$$\begin{aligned} m-n &= 2-3 \\ &= -1. \end{aligned}$$